

Solution Key to Problem Sheet 2

Ques1.

a) S1: $y_1(t) = 2x_1(2t+2)$

This is invertible function, so we can find inverse

$$x_1(t) = 2y_1(2t+2)$$

Let $2t+2 = z$, then

$$x_1((z-2)/2) = 2y_1(z)$$

Now change the variable

$$y_1(t) = \frac{1}{2} x_1((t-2)/2) : S2$$

- b) The system is not invertible as time scaling with factor more than 1 leads to decimation of samples.
- c) The inverse system is non-causal so this system is realizable only for offline processing.

Ques 2:

- a) i) If the system is additive, the following relation is true.

$$\begin{aligned} x_1(t) + x_2(t) &= y_1(t) + y_2(t) \\ 0 &= 0 \end{aligned}$$

Now if the system is homogenous, we proceed as follows:

$$ax(t) = ay(t)$$

$$\text{If } x(t) = 0 \text{ then } y(t) = 0$$

If $a = 0$, zero-input gives zero-output

- b) $y(t) = \sin\{x(t)\}$
- c) No. Consider $x(t) = 0$, for $t < 0$

1, otherwise

And $y(t) = x(t+2)$. Then $y(t)$ is not zero for $t < 0$

Ques3.

a) Let $x(t)$ gives $y(t)$

If system is time-invariant then $x(t+T)$ gives $y(t+T)$

Also, $x(t) = x(t+T)$ [being periodic]

implies $x(t)$ gives $y(t+T)$. But $x(t)$ gives $y(t)$ as assumed before

This implies $y(t) = y(t+T)$

Similarly, let $x[n]$ give $y[n]$ as output. Let $x[n]$ be periodic with period N . This implies that $x[n+N] = x[n]$. Also since the system is LTI, $x[n+N]$ gives $y[n+N]$. This implies $x[n]$ gives $y[n+N]$. However since we had assumed that $x[n]$ gives $y[n]$, then $y[n] = y[n+N]$.

b) Consider the time invariant system $y(t) = \sin[x(t)]$. Let $x(t) = t$ (non-periodic). Then $y(t) = \sin(t)$, which is a periodic signal.

Ques4.

Assumption: if $x(t)=0$ for $t < t_0$ then $y(t)=0$ for $t < t_0$

To prove that : the system is causal.

Let us consider an arbitrary signal $x_1(t)$. then , let us consider another signal $x_2(t)$ which is same as $x_1(t)$ for $t < t_0$

Since the system is linear,

$$x_1(t) - x_2(t) \rightarrow y_1(t) - y_2(t)$$

Since $x_1(t) - x_2(t)=0$ for $t < t_0$, by our assumption $y_1(t) - y_2(t) = 0$ for $t < t_0$. This implies $y_1(t) = y_2(t)$ for $t < t_0$ which is property of the causal system.

Assumption: The system is causal.

To prove that: If $x(t) = 0$ for $t < t_0$, then $y(t) = 0$ for $t < t_0$.

Let us assume that the signal $x(t) = 0$ for $t < t_0$. Then we may express $x(t)$ as $x(t) =$

$x_1(t) - x_2(t)$, where $x_1(t) = x_2(t)$ for $t < t_0$. Since the system is linear, the output to $x(t)$ will be $y(t) = y_1(t) - y_2(t)$. Now, since the system is causal, $x_1(t) = x_2(t)$ for $t < t_0$ implies that $y_1(t) = y_2(t)$ for $t < t_0$. Therefore, $y(t) = 0$ for $t < t_0$

b) $y(t)=x(t)x(t+5)$; now $x(t)=0$ for $t < t_0$ implies that $y(t)=0$ for $t < t_0$.

Note that the system is non-linear and non-causal.

c) $y(t)=x(t)+10$. This system is non-linear and causal. This does not satisfy the condition of part (a)

d) Let us assume that $x[n] \neq 0$ gives $y[n] = 0$

Then by linearity $ax[n] \rightarrow 0$, which implies that for distinct values of a , we will still get the same output 0, which contradicts with the definition of invertibility that the distinct inputs should lead to distinct outputs.

e) $y[n]=x^2[n]$

Ques.5

a) Linear, Time variant and Non causal

$$\varphi_{hx}(t) = \int_{-\infty}^{\infty} h(t+\tau)x(\tau) d\tau$$

Replacing $x(t)$ by $cx(t)$ where c is a real number

$$\tilde{\varphi}_{hx}(t) = \int_{-\infty}^{\infty} h(t+\tau)cx(\tau) d\tau = c\varphi_{hx}(t)$$

Similarly putting $\tilde{x}(t) = x_1(t) + x_2(t)$ we will get $\tilde{\varphi}_{hx}(t) = \varphi_{hx_1}(t) + \varphi_{hx_2}(t)$

Thus $\varphi_{hx}(t)$ obeys homogeneity and additivity. That's implies $\varphi_{hx}(t)$ is linear.

Now substitute $x(t)$ by $x(t - t_0)$

$$\tilde{\varphi}_{hx}(t) = \int_{-\infty}^{\infty} h(t+\tau)x(\tau - t_0) d\tau$$

Let $\tau - t_0 = m$,

so $\tau = m + t_0$ and $d\tau = dm$

Which can be arranged to

$$\text{LHS} = \int_{-\infty}^{\infty} h(t+t_0+m)x(m) dm = \varphi_{hx}(t+t_0) \neq \varphi_{hx}(t-t_0)$$

Hence the system is time variant.

As the system output depends upon the values of input from $-\infty$ to $+\infty$, the system is non-casual.

b) Linear, Time-invariant and Non-causal.

Ques.6

- a) S is an incrementally linear system, so the system 'S' is an addition of linear system 'L' and zero input response.

The overall response of the system

$$= (\text{the response of the system 'S' to } \{x[n] + x_1[n]\}) - y_1[n]$$

$$= (\text{the response of the system 'S' to } \{x[n] + x_1[n]\}) - \text{the response of the system 'S' to } x_1[n]$$

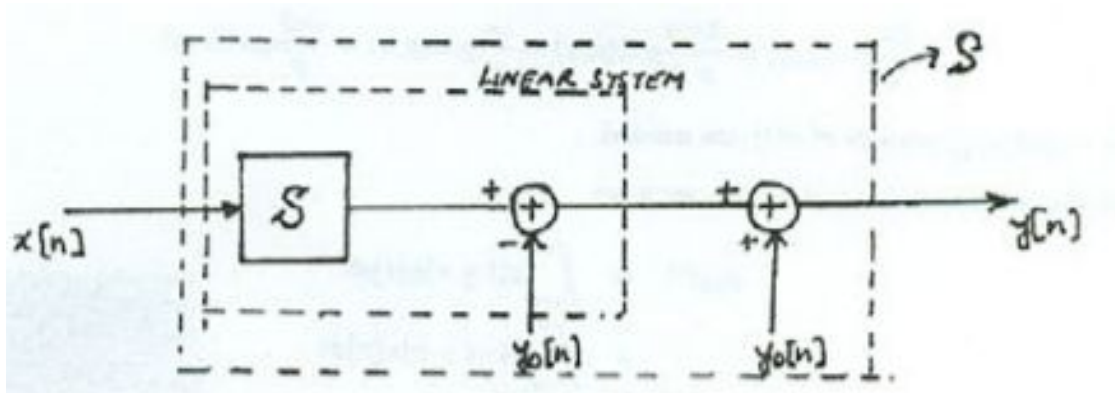
$$= (\text{response of linear system L to } \{x[n] + x_1[n]\} + \text{zero input response of S}) - (\text{response of a linear system L to } x_1[n] + \text{zero input response of S})$$

$$= \text{Response of a linear system L to } x[n]$$

So, $y[n]$ does not depend on the particular choice of $x_1[n]$.

- b) From (a), $y[n]$ does not depend on $x_1[n]$. So, we can assume it to be zero.

If $x_1[n] = 0$ for all n , then $y_1[n]$ will be the zero input response $y_0[n]$. S may then be redrawn as shown in below



- c) i) Incrementally linear

$$x[n] \rightarrow x[n] + 2x[n+4] \quad \text{and} \quad y_0[n] = n$$

- ii) Incrementally linear, where

$$x[n] = 0 \quad \text{for } n \text{ even}$$

$$x[n] = \sum_{k=-\infty}^{(n-1)/2} x[k] \quad \text{for } n \text{ odd}$$

and

$$y_0[n] = \frac{n}{2} \quad \text{for } n \text{ even}$$

$$= \frac{n-1}{2} \quad \text{for } n \text{ odd}$$

Ques.7

(a) $y(t) = x(t - 2) + x(2 - t)$:

Memory: as time shift is involved

Time invariance: Yes

Linear: True

Non causal: $x(2-t)$ is not causal

Stable: True BIBO

(b) $y(t) = [\cos(3t)]x(t)$

Memory less: True

Time variant: function of $\cos(t)$ hence time variant

Linear: Yes

Causal: dependent on t only

Stable: at all points $y(t) < x(t)$ so stable as BIBO stability

(c) Memory: Integral from $-\infty$ hence depends on past

Time variant: integral limit is $2t$

Linear: Integral is linear operation

Non causal: Integral to $2t$

Unstable: Integral of $u(t)$

(d) Memory, Time variant, Linear, Non causal, Unstable

(e) Memory, Time variant, Linear, Causal, Stable

(f) Memory, Time Invariant, Linear, Causal, Stable

(g) Memory, Time variant, Linear, Non causal, Stable

(h) Memoryless, Time invariant, Linear, Non causal, Unstable

(i) Memory, Time variant, Linear, Causal, Unstable

- (j) Memory, Time Invariant, Nonlinear, Causal, Stable
- (k) Memory, Time Invariant, Linear, Non-Causal, Stable
- (l) Memory, Time variant, Linear, Non Causal, Stable
- (m) Memory, Time variant, Linear, Non causal, Stable
- (n) Memory, Time Invariant, Linear, Non causal, Stable